Variance and Central Limit Theorem for the Log Coprime Density of Random LCMs

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Abstract

Let $X_1, X_2, \dots$ be independent and identically distributed random variables uniform on $\{2, 3, \dots, M\}$, let $L_k = \operatorname{lcm}(X_1, \dots, X_k)$, and let

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E}\,\ln(\varphi(L_k)/L_k)$$

denote the centred log coprime density, with $s_k^2 := \operatorname{Var} W_k(M)$. A companion paper [15] established an exact subset identity for $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$, a sharp asymptotic equivalence $\rho_k(M) \to P_M$ at fixed M, and the joint-regime convergence $\ln M \cdot (\rho_{\lfloor cM \rfloor}(M) - P_M)/P_M \to A(c)$ with $A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j)$. We study second-order fluctuations in the same joint regime $k = \lfloor cM \rfloor$, $M \to \infty$, with fixed $c > 0$. We prove three results. First, the variance asymptotic

$$M \ln M \cdot s_k^2 \longrightarrow B(c) := \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1},$$

via a diagonal / off-diagonal decomposition where the off-diagonal term is shown to be negligible by an exact covariance identity valid for prime pairs (p, q) with $pq > M$. Second, a central limit theorem: $W_k(M)/s_k \xrightarrow{d} \mathcal{N}(0, 1)$ as $M \to \infty$ in the joint regime. The CLT proof uses the Hall–Heyde martingale central limit theorem applied to the Doob filtration $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$. Two ingredients enter. A uniform increment bound $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$ on a super-polynomially likely event combines a coupon-collector tail estimate with a sieve-geometric observation ($(M/(\log M)^2)^2 > M$ for M large), either fact alone being insufficient. Conditional-variance concentration is obtained from explicit moment bounds for the hitting-time variables $T_p := \min\{i : p \mid X_i\}$: a per-prime variance bound $\operatorname{Var}(R_{pp}^{(k)}) \leq 3q_p^k/\pi_p^2$ and a cross-prime covariance bound $|\operatorname{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c) k q_p^k q_q^k$ valid when $pq > M$ and $\pi_p + \pi_q \leq 1/3$, with an explicit constant $C_0(c)$ depending only on c, together with an L^2 Minkowski summation that controls the covariances among all distinct prime-pair terms. Third, a Berry–Esseen bound: the Kolmogorov distance between the law of $W_k(M)/s_k$ and the standard normal is $O_c((\ln M)^{-2/5})$, in both the logarithmic and the linear scale. Its proof combines the Heyde–Brown martingale inequality with a quantitative refinement of the variance asymptotic, $M \ln M \cdot s_k^2 = B(c) + O_c(1/\ln M)$, obtained from the effective prime-counting bounds of Rosser and Schoenfeld.

Keywords: Random LCM, Euler totient, coprime density, central limit theorem, martingale central limit theorem, Berry–Esseen bound, Doob decomposition, Hall–Heyde, probabilistic number theory.

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1 Introduction

The arithmetic structure of the least common multiple of a random set of integers has been studied extensively in the regime where the support size n grows while the number of samples k stays fixed or bounded; see Cilleruelo, Rué, Šarka, and Zumalacárregui [4]; Hilberdink and Tóth [7]; Alsmeyer, Kabluchko, and Marynych [1]; Bostan, Marynych, and Raschel [2]; Buraczewski, Iksanov, and Marynych [3]; Iksanov, Marynych, and Raschel [8]; Kabluchko, Marynych, and Raschel [9]; and Sanna [13]. These works establish limit theorems for $\log \operatorname{lcm}$ and for multiplicative functions of the LCM in the fixed-k, $n \to \infty$ regime.

A complementary regime, in which the support $\{2, \ldots, M\}$ is fixed and the sample size $k \to \infty$, was introduced in the companion paper [15]. There the natural object of study is not the LCM itself but the coprime density

$$\frac{\varphi(L_k)}{L_k} = \prod_{p \mid L_k} \left(1 - \frac{1}{p}\right),$$

whose large-k behaviour is governed by exactly which primes $p \le M$ appear as divisors of $L_k = \operatorname{lcm}(X_1, \ldots, X_k)$. The paper [15] established, writing $\rho_k(M) := \mathbb{E}[\varphi(L_k)/L_k]$, $P_M := \prod_{p \le M}(1 - 1/p)$, $m_p := \lfloor M/p \rfloor$, $\pi_p := m_p/(M-1)$, $q_p := 1 - \pi_p$:

- *Exact identity:*
$$\rho_k(M) = P_M \sum_{T \subseteq \mathcal{P}_M} \beta_T(M)^k \prod_{p \in T} (p-1)^{-1},$$
where $\beta_T(M) = \mathbb{P}(\gcd(X, \prod_{p \in T} p) = 1)$ for a single uniform draw X.
- *Fixed-M limit with sharp rate:* $\rho_k(M) \to P_M$ as $k \to \infty$, with an explicit geometric rate of convergence controlled by $(q^{**}(M)/q^*(M))^k$.
- *Joint-regime first-order asymptotic:* in the regime $k = \lfloor cM \rfloor$, $M \to \infty$, fixed $c > 0$,
$$\ln M \cdot \frac{\rho_k(M) - P_M}{P_M} \longrightarrow A(c) := \sum_{j \ge 1} e^{-cj} \ln(1 + 1/j).$$

Dependence on the companion paper. The proofs below are self-contained apart from three explicitly identified inputs from [15]: the subset-density bounds collected in Lemma 4, the level-count estimate quoted in Lemma 5, and the elementary disjointness observation quoted in Proposition 6. The exact identity and first-order asymptotic for $\rho_k(M)$ are recalled only to situate the present second-order results and are not used in the proofs of Theorems 1 and 2. The disjointness identity needed later is also rederived directly in Lemma 8.

In this paper we study the second-order behaviour of the centred log coprime density

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E}\, \ln(\varphi(L_k)/L_k),$$

in the same joint regime $k = \lfloor cM \rfloor$, $M \to \infty$, with fixed $c > 0$. We prove three results.

Theorem 1 (Variance asymptotic). *For each fixed $c > 0$, in the joint regime $k = \lfloor cM \rfloor$, $M \to \infty$,*

$$M \ln M \cdot s_k^2 \;\longrightarrow\; B(c) := \sum_{j \ge 1} e^{-cj}(1 - e^{-cj}) = \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1}, \tag{1}$$

where $s_k^2 = \operatorname{Var} W_k(M)$.

Theorem 2 (Central limit theorem)**.** *For each fixed $c > 0$, in the joint regime $k = \lfloor cM \rfloor$, $M \to \infty$,*

$$\frac{W_k(M)}{s_k} \xrightarrow{d} \mathcal{N}(0,1).$$

Consequently,

$$\sqrt{M \ln M} \cdot \frac{W_k(M)}{\sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0,1), \qquad \sqrt{M \ln M} \cdot \frac{\varphi(L_k)/L_k - \rho_k(M)}{P_M \sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0,1).$$

The first consequence follows from Theorem 1 and Slutsky's theorem; the second follows from the linear-scale transfer Lemma 20 below.

Theorem 3 (Berry–Esseen rate)**.** *For each fixed $c > 0$ there exist $C(c)$ and $M_0(c)$ such that, for all $M \geq M_0(c)$ and $k = \lfloor cM \rfloor$,*

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}\left(\frac{W_k(M)}{s_k} \leq x \right) - \Phi(x) \right| \leq \frac{C(c)}{(\ln M)^{2/5}},$$

where Φ denotes the standard normal distribution function. The same bound, with a possibly larger $C(c)$, holds for the renormalised statistics $\sqrt{M \ln M}\, W_k(M)/\sqrt{B(c)}$ and $\sqrt{M \ln M}\, \big(\varphi(L_k)/L_k - \rho_k(M)\big)/\big(P_M \sqrt{B(c)}\big)$.

The rate in Theorem 3 is logarithmic because it inherits the L^2 concentration scale of the predictable quadratic variation established in Theorem 19; Section 10 identifies precisely what would be needed to reach a power of M.

1.1 Method and outline

Theorem 1 is proved by a direct diagonal / off-diagonal decomposition. Writing $X_p = \mathbf{1}\{p \mid L_k\}$ so that $\ln(\varphi(L_k)/L_k) = \sum_{p \leq M} X_p \ln(1 - 1/p)$, the variance decomposes as

$$s_k^2 = V_1(k, M) + V_2(k, M)$$

with V_1 the diagonal variance contribution and V_2 the off-diagonal covariance sum. The diagonal V_1 is handled by partitioning primes by level $m_p = j$ and applying a partial-summation analogue of Mertens' second theorem that yields $\sum_{p:\, m_p = j} 1/p^2 \sim 1/(M \ln M)$ for each fixed j, giving $M \ln M \cdot V_1 \to B(c)$ by dominated convergence. The off-diagonal V_2 is bounded using the exact covariance identity

$$\beta_{\{p,q\}}(M) - q_p q_q = -\frac{m_p m_q}{(M-1)^2} \quad \text{for } pq > M,$$

together with a mean-value bound. Section 3 carries this out in detail.

Theorem 2 is proved via the Hall–Heyde martingale central limit theorem [5, Thm. 3.2], applied to the Doob filtration $\mathcal{F}_i = \sigma(X_1, \ldots, X_i)$. Section 4 derives the martingale decomposition $W_k = \sum_{i=1}^{k} D_i$ and identifies the explicit form of the increments.

The application of [5] requires verifying two conditions. Section 5 proves the conditional Lindeberg condition via a uniform bound on the increments: $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$ on a super-polynomially likely event. The bound exploits two facts simultaneously. First, a coupon-collector tail estimate: with super-polynomially high probability, every prime p with $m_p \geq (\log M)^2$ has been sampled by time $k/2$. Second, a sieve-geometric observation: once contributing primes are restricted to $p > M/(\log M)^2$, at most one of them can divide any given

$x \le M$, because $(M/(\log M)^2)^2 > M$ for M sufficiently large. Either fact alone gives only a weak bound; together they yield the required estimate.

Section 6 proves the conditional-variance concentration $\sum_{i=1}^{k} \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]/s_k^2 \xrightarrow{P} 1$. The argument rests on two explicit moment bounds. Lemma 16 controls the per-prime variance $\mathrm{Var}(R_{pp}^{(k)})$ via a substitution $Y_p = q_p^{-2(T_p \wedge k)}$ that reduces the quantity to a function of a single geometric random variable. Lemma 17 bounds the cross-prime covariance $|\,\mathrm{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})|$ for $pq > M$ via a comparison between the true joint law of the hitting times (T_p, T_q) and its independent product. These bounds are combined with a single-pair second-moment estimate, a level-summation argument, and Minkowski's inequality in L^2, which controls all cross-covariances among the prime-pair summands without assuming orthogonality.

Section 7 assembles the pieces for Theorem 2. Section 8 upgrades Theorem 2 to the Berry–Esseen bound of Theorem 3 via the Heyde–Brown martingale inequality [6]; the required inputs are a fourth-moment estimate derived from Theorem 14, the conditional-variance bound of Theorem 19, and a quantitative refinement of Theorem 1 (Proposition 12, Section 3.3) that replaces the pointwise prime number theorem by the effective bounds of Rosser and Schoenfeld [12]. Section 9 presents numerical diagnostics for the theorems.

2 Preliminaries

Throughout, $M \ge 2$ is a fixed integer, $X_1, X_2, \ldots$ are i.i.d. uniform on $\{2, \ldots, M\}$, $L_k = \mathrm{lcm}(X_1, \ldots, X_k)$, and $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$. For a prime $p \le M$, set

$$m_p := \lfloor M/p \rfloor, \quad \pi_p := \frac{m_p}{M-1}, \quad q_p := 1 - \pi_p, \quad \xi_p^{(i)} := \mathbf{1}\{p \mid X_i\} - \pi_p.$$

Then m_p is the number of multiples of p in $\{2, \ldots, M\}$ and $\pi_p = \mathbb{P}(p \mid X_i)$ for any single draw. For $T \subseteq \mathcal{P}_M := \{p : p \le M\}$, let $A_T(M) = \{n \in \{2, \ldots, M\} : \gcd(n, \prod_{p \in T} p) = 1\}$, so $\beta_T(M) = |A_T(M)|/(M-1)$ is the probability that a single uniform draw is coprime to every prime in T; in particular $\beta_\varnothing = 1$ and $\beta_{\{p\}} = q_p$. Let $T_p := \min\{i \ge 1 : p \mid X_i\}$, which is geometric with success probability π_p. Let $\mathcal{F}_i := \sigma(X_1, \ldots, X_i)$ and $\mathcal{F}_0 := \{\varnothing, \Omega\}$. Logarithms are natural throughout, and $c > 0$ is fixed with $k := \lfloor cM \rfloor$ unless otherwise stated. We write C for an absolute constant and $C(c)$ for a constant depending only on c; these may vary from line to line.

We record three elementary facts and the basic subset-density lemma.

Lemma 4 (Elementary subset-density bounds; [15, Lems. 2.1–2.4])**.** *For every $T \subseteq \mathcal{P}_M$:*

(i) $\mathbb{P}(\textit{no prime in } T \textit{ divides } L_k) = \beta_T(M)^k$;

(ii) $\beta_T(M) \le q_p(M)$ *for every* $p \in T$, $T \ne \varnothing$;

(iii) if $|T| = r \ge 1$, *then* $\beta_T(M)^k \le \prod_{p \in T} q_p(M)^{k/r}$ *for every real* $k > 0$.

Lemma 5 (Rosser–Schoenfeld level envelope; [12], [15, Lem. 6.3])**.** *For all $M \ge 289$ and all $1 \le j \le M/17$,*

$$\left|\{p \le M : m_p = j\}\right| \le \frac{1.26\,(M/j)}{\ln(M/j)}, \qquad \sum_{p:\, m_p = j} \frac{1}{p-1} \le \frac{5.04}{\ln(M/j)}.$$

In particular, for $1 \le j \le M^{1/2}$,

$$\left|\{p \le M : m_p = j\}\right| \le \frac{2.52\,M}{j \ln M}, \qquad \sum_{p:\, m_p = j} \frac{1}{p-1} \le \frac{10.08}{\ln M}.$$

Proposition 6 (Disjointness for $pq > M$; [15, Rem. 5.6]). *For distinct primes $p, q \le M$ with $pq > M$, the events $\{p \mid X\}$ and $\{q \mid X\}$ are disjoint under a single uniform draw X on $\{2, \dots, M\}$; equivalently $\mathbb{P}(p \mid X, q \mid X) = 0$, so $\mathbb{P}(p \nmid X, q \nmid X) = 1 - \pi_p - \pi_q$ and*

$$\beta_{\{p,q\}}(M) - q_p(M)\, q_q(M) = -\pi_p \pi_q.$$

Part (iii) of Lemma 4 is the geometric-mean bound of [15, Lem. 2.4], which underlies the sharp joint-scaling result in the companion paper.

3 Variance asymptotic: proof of Theorem 1

Let $k = \lfloor cM \rfloor$. Write $\varphi(L_k)/L_k = \prod_{p \le M}(1 - 1/p)^{X_p}$, where $X_p = \mathbf{1}\{p \mid L_k\}$, so

$$\ln\bigl(\varphi(L_k)/L_k\bigr) = \sum_{p \le M} X_p \ln(1 - 1/p).$$

The variance of X_p is $(1 - q_p^k) q_p^k$ and for $p \neq q$,

$$\mathrm{Cov}(X_p, X_q) = \mathbb{P}(p \mid L_k,\ q \mid L_k) - (1 - q_p^k)(1 - q_q^k) = \beta^k_{\{p,q\}} - q_p^k q_q^k,$$

as follows from the inclusion-exclusion identity $\mathbb{P}(p \mid L_k, q \mid L_k) = 1 - q_p^k - q_q^k + \beta^k_{\{p,q\}}$. Therefore

$$s_k^2 = V_1(k, M) + V_2(k, M), \tag{2}$$

where the diagonal and off-diagonal contributions are

$$\begin{aligned} V_1(k, M) &= \sum_{p \le M} \ln(1 - 1/p)^2\, q_p^k (1 - q_p^k), \\ V_2(k, M) &= \sum_{p \neq q,\ p,q \le M} \ln(1 - 1/p) \ln(1 - 1/q) \bigl(\beta^k_{\{p,q\}} - q_p^k q_q^k\bigr). \end{aligned}$$

3.1 Diagonal contribution

We show $M \ln M \cdot V_1(k, M) \to B(c)$. Partition by $m_p = j$:

$$V_1(k, M) = \sum_{j \ge 1} \bigl(1 - j/(M-1)\bigr)^k \bigl(1 - (1 - j/(M-1))^k\bigr) \sum_{p\colon m_p = j} \ln(1 - 1/p)^2.$$

For $p \in (M/(j+1), M/j]$, $\ln(1 - 1/p)^2 = 1/p^2 + O(1/p^3)$, and since $p \ge M/(j+1)$ we have $1/p^3 = O((j+1)^3/M^3)$. Hence

$$\sum_{p\colon m_p = j} \ln(1 - 1/p)^2 = \sum_{p\colon m_p = j} \frac{1}{p^2} + O\left(\frac{(j+1)^3}{M^3} \cdot \pi(M/j) \right) = \sum_{p\colon m_p = j} \frac{1}{p^2} + O\left(\frac{(j+1)^2}{M^2 \ln M} \right).$$

Lemma 7 (Level-j sum of reciprocal squares). *For each fixed $j \ge 1$,*

$$M \ln M \cdot \sum_{p\colon m_p = j} \frac{1}{p^2} \longrightarrow 1 \qquad (M \to \infty).$$

Proof. By partial summation and the prime number theorem in its standard quantitative form [14], $\sum_{p\le x} 1/p^2 = \pi(x)/x^2 + 2\int_2^x \pi(t)/t^3\,dt$, so $\sum_{p\in(A,B]} 1/p^2 = \pi(B)/B^2 - \pi(A)/A^2 + 2\int_A^B \pi(t)/t^3\,dt$. With $A = M/(j+1)$, $B = M/j$, and using $\pi(t) = t/\ln t + O(t/(\ln t)^2)$,

$$\pi(B)/B^2 = \frac{1}{B\ln B}(1+O(1/\ln B)) = \frac{j}{M\ln(M/j)}(1+o(1)) = \frac{j}{M\ln M}(1+o(1)),$$

and analogously $\pi(A)/A^2 = (j+1)/(M\ln M)(1+o(1))$. Their difference is

$$\pi(B)/B^2 - \pi(A)/A^2 = -\frac{1}{M\ln M}(1+o(1)).$$

The integral is

$$2\int_A^B \frac{1}{t^2\ln t}(1+O(1/\ln t))\,dt = \frac{2}{\ln M}\int_A^B \frac{dt}{t^2}(1+o(1)) = \frac{2}{\ln M}\left(\frac{1}{A}-\frac{1}{B}\right)(1+o(1)),$$

and $1/A - 1/B = (j+1)/M - j/M = 1/M$, so the integral contributes $2/(M\ln M)(1+o(1))$. Summing the two contributions:

$$\sum_{p:\, m_p=j} \frac{1}{p^2} = \left[-\frac{1}{M\ln M} + \frac{2}{M\ln M}\right](1+o(1)) = \frac{1}{M\ln M}(1+o(1)),$$

for each fixed $j \ge 1$. □

Dominated convergence. For each fixed j, Lemma 7 together with $(1-j/(M-1))^k \to e^{-cj}$ and $(1-(1-j/(M-1))^k) \to 1-e^{-cj}$ gives

$$M\ln M \cdot (1-j/(M-1))^k\big(1-(1-j/(M-1))^k\big) \sum_{p:\, m_p=j} \ln(1-1/p)^2 \longrightarrow e^{-cj}(1-e^{-cj}).$$

A uniform envelope follows directly from the prime-count part of Lemma 5, applied to the summand $\ln(1-1/p)^2$ itself rather than to its leading term $1/p^2$. If $m_p = j$, then $p > M/(j+1)$, so $p-1 > M/(2(j+1))$ whenever $M \ge 2(j+1)$, and hence $\ln(1-1/p)^2 \le (p-1)^{-2} < 4(j+1)^2/M^2$. Therefore, for $1 \le j \le M^{1/2}$ and $M \ge 289$ (which guarantees both $\ln(M/j) \ge \frac{1}{2}\ln M$ and $M \ge 2(j+1)$ in this range),

$$\sum_{p:\, m_p=j} \ln(1-1/p)^2 \le \big|\{p : m_p = j\}\big| \frac{4(j+1)^2}{M^2} \le \frac{2.52\,M}{j\ln M}\cdot\frac{4(j+1)^2}{M^2} = \frac{10.08\,(j+1)^2}{Mj\ln M}.$$

For the exponential factor, since $k \ge cM-1$ and $1-x \le e^{-x}$,

$$(1-j/(M-1))^k \le e^{-kj/(M-1)} \le e^{-cj}\,e^{j/(M-1)} \le 2e^{-cj} \qquad (1 \le j \le M^{1/2},\ M \ge 5).$$

Hence each level-j term of $M\ln M \cdot V_1$ is bounded by $10.08\,(j+1)^2/j \cdot 2e^{-cj}$, which is summable in j. The tail $j > M^{1/2}$ decays super-exponentially, since there $(1-j/(M-1))^k \le e^{-c\sqrt{M}/2}$ for $M \ge M_0(c)$ while $\sum_{p\le M} \ln(1-1/p)^2 \le \sum_{n\ge 2}(n-1)^{-2} = O(1)$. By dominated convergence,

$$\lim_{M\to\infty} M\ln M \cdot V_1(k,M) = \sum_{j\ge 1} e^{-cj}(1-e^{-cj}) = B(c). \tag{3}$$

3.2 Off-diagonal contribution is negligible

We show $M \ln M \cdot V_2(k, M) \to 0$. The starting point is an exact identity valid in a restricted range of prime pairs.

Lemma 8 (Covariance identity for disjoint prime levels). *Let $p, q \in \mathcal{P}_M$ with $p \neq q$ and $pq > M$. Then*

$$\beta_{\{p,q\}}(M) - q_p(M)\, q_q(M) = -\frac{m_p\, m_q}{(M-1)^2}. \tag{4}$$

Proof. By inclusion-exclusion, $|A_{\{p,q\}}(M)| = M - \lfloor M/p \rfloor - \lfloor M/q \rfloor + \lfloor M/(pq) \rfloor - 1$. When $pq > M$, $\lfloor M/(pq) \rfloor = 0$, so $(M-1)\beta_{\{p,q\}} = M - 1 - m_p - m_q$. Hence

$$(M-1)^2\big(\beta_{\{p,q\}} - q_p q_q\big) = (M-1)(M-1-m_p-m_q) - (M-1-m_p)(M-1-m_q) = -m_p m_q. \ \square$$

Remark 9. For levels (j_1, j_2) with $(j_1+1)(j_2+1) < M$, every prime p with $m_p = j_1$ and q with $m_q = j_2$ satisfies $pq \geq M/(j_1+1) \cdot M/(j_2+1) = M^2/((j_1+1)(j_2+1)) > M$, so (4) applies. In particular, if $\max(j_1, j_2) \leq M^{1/4}$ then $(j_1+1)(j_2+1) \leq (M^{1/4}+1)^2 < M$ for all $M \geq 7$, so the identity applies to every pair in this range.

Main-range bound. Let $\mathcal{J}(M) = \{(p,q) : p \neq q,\ p, q \leq M,\ \max(m_p, m_q) \leq M^{1/4}\}$ and write

$$V_2^{\text{main}}(k, M) := \sum_{(p,q) \in \mathcal{J}(M)} \ln(1-1/p) \ln(1-1/q) \big(\beta^k_{\{p,q\}} - q_p^k q_q^k\big),$$

with $V_2^{\text{tail}}(k, M) := V_2(k, M) - V_2^{\text{main}}(k, M)$.

For $(p,q) \in \mathcal{J}(M)$, let $j_1 = m_p$, $j_2 = m_q$. By Remark 9 and Lemma 8, $\beta_{\{p,q\}} - q_p q_q = -j_1 j_2/(M-1)^2$; in particular $\beta_{\{p,q\}} < q_p q_q$, so by the mean-value form $|a^k - b^k| \leq k|a-b| \max(a,b)^{k-1}$ with $a = \beta_{\{p,q\}}, b = q_p q_q$,

$$\big|\beta^k_{\{p,q\}} - q_p^k q_q^k\big| \leq \frac{k\, j_1 j_2}{(M-1)^2}\, (q_p q_q)^{k-1}. \tag{5}$$

Since $q_p q_q \geq 1 - (j_1+j_2)/(M-1) \geq 1/2$ for M large enough (because $j_1 + j_2 \leq 2M^{1/4} = o(M)$), we have $(q_p q_q)^{k-1} \leq 2(q_p q_q)^k$. Using also $|\ln(1-1/p)| \leq 1/(p-1)$, the contribution of any pair $(p,q) \in \mathcal{J}(M)$ with $m_p = j_1, m_q = j_2$ is at most

$$\frac{1}{(p-1)(q-1)} \cdot \frac{2k\, j_1 j_2}{(M-1)^2}\, (q_p q_q)^k.$$

Summing over pairs at fixed levels (j_1, j_2) — noting $(q_p q_q)^k$ depends on the pair only through j_1, j_2 — and using $S_j(M) := \sum_{p:\, m_p = j} 1/(p-1)$,

$$\sum_{\substack{(p,q) \in \mathcal{J}(M) \\ m_p = j_1,\, m_q = j_2}} \frac{1}{(p-1)(q-1)} \leq S_{j_1}(M)\, S_{j_2}(M).$$

By Lemma 5, $S_j(M) \leq 10.08/\ln M$ uniformly for $1 \leq j \leq M^{1/2}$ and $M \geq 289$, hence certainly for $j \leq M^{1/4}$. Moreover, $(q_p q_q)^k \leq 2\, e^{-c(j_1+j_2)}$ uniformly over $j_1, j_2 \leq M^{1/4}$ once $M \geq M_0(c)$: indeed $\ln(1-x) = -x + O(x^2)$ and $k/(M-1) = c + O(1/M)$ give

$$k \ln(q_p q_q) = -\frac{k(j_1+j_2)}{M-1} + O\Big(\frac{k(j_1^2+j_2^2)}{M^2}\Big) = -c(j_1+j_2) + O\big(M^{-1/2}\big), \tag{6}$$

uniformly in this range, since $j_1^2 + j_2^2 \le 2M^{1/2}$. Therefore

$$|V_2^{\mathrm{main}}(k, M)| \le \frac{4k}{(M-1)^2} \cdot \frac{(10.08)^2}{(\ln M)^2} \cdot \sum_{j_1, j_2 \ge 1} j_1 j_2 \, e^{-c(j_1+j_2)}.$$

The double sum factorises:

$$\sum_{j_1, j_2 \ge 1} j_1 j_2 \, e^{-c(j_1+j_2)} = \left(\sum_{j \ge 1} j \, e^{-cj} \right)^2 = \frac{e^{2c}}{(e^c - 1)^4}.$$

With $k = \lfloor cM \rfloor$ so $k/(M-1)^2 = O(1/M)$,

$$|V_2^{\mathrm{main}}(k, M)| = O\left(\frac{1}{M (\ln M)^2} \right). \tag{7}$$

Tail-range bound. For $(p, q) \notin \mathcal{J}(M)$, at least one of m_p, m_q exceeds $M^{1/4}$. Say $m_p > M^{1/4}$; then $q_p^k \le (1 - M^{1/4}/(M-1))^{\lfloor cM \rfloor} \le \exp(-cM^{1/4}(1 + o(1)))$. Using Lemma 4(ii) in the form $\beta_{\{p,q\}} \le q_p$, $\beta_{\{p,q\}}^k \le q_p^k$, and the trivial bound $q_p^k q_q^k \le q_p^k$,

$$\left| \beta_{\{p,q\}}^k - q_p^k q_q^k \right| \le 2 \, q_p^k.$$

Writing $\Lambda(M) := \sum_{p \le M} 1/(p-1) = O(\ln \ln M)$ (Mertens),

$$\begin{aligned}
|V_2^{\mathrm{tail}}(k, M)| &\le 2 \sum_{p \colon m_p > M^{1/4}} \frac{q_p^k}{p-1} \sum_{q \le M,\, q \ne p} \frac{1}{q-1} + (\text{symmetric term with } p, q \text{ swapped}) \\
&\le 4 \, \Lambda(M) \, \exp(-cM^{1/4}(1 + o(1))) \, \Lambda(M) \\
&= O\big((\ln \ln M)^2 \, e^{-cM^{1/4}} \big),
\end{aligned}$$

which vanishes super-polynomially in M. In particular $M \ln M \cdot V_2^{\mathrm{tail}} \to 0$.

Combining (7) and the tail bound,

$$|V_2(k, M)| = O\left(\frac{1}{M (\ln M)^2} \right), \tag{8}$$

so $M \ln M \cdot |V_2(k, M)| = O(1/\ln M) \to 0$. Combined with (3), this gives (1). $\square$

Remark 10 (Sign of V_2). The leading contribution to V_2 comes from the region $\max(m_p, m_q) \le M^{1/4}$. There $\beta_{\{p,q\}} - q_p q_q = -m_p m_q/(M-1)^2 < 0$, so $\beta_{\{p,q\}}^k - q_p^k q_q^k < 0$. The weights $\ln(1 - 1/p) \ln(1 - 1/q)$ are positive. Hence the main-region sum, and therefore V_2 for M large, is strictly negative. Numerically (Section 9) V_2 is observed negative at every tested (k, M).

Remark 11 ($s_k \asymp (M \ln M)^{-1/2}$). Theorem 1 gives $s_k^2 \sim B(c)/(M \ln M)$ with $B(c) > 0$ for every $c > 0$, so in particular $s_k \asymp (M \ln M)^{-1/2}$ in the joint regime. This scale is used throughout the CLT proof below.

3.3 A quantitative refinement

The dominated-convergence argument of Section 3.1 yields convergence without a rate. The Berry–Esseen bound of Section 8 requires the following quantitative version, in which the pointwise prime number theorem used in Lemma 7 is replaced by the effective bounds of Rosser and Schoenfeld.

Proposition 12 (Quantitative variance asymptotic)**.** *For each fixed $c > 0$ there exist $C(c)$ and $M_0(c)$ such that, for all $M \geq M_0(c)$ and $k = \lfloor cM \rfloor$,*

$$\left| M \ln M \cdot s_k^2 - B(c) \right| \leq \frac{C(c)}{\ln M}. \tag{9}$$

Proof. By (8), $M \ln M \cdot |V_2| = O(1/\ln M)$, so it suffices to prove (9) with V_1 in place of s_k^2.

Step 1: effective level estimate. Rosser and Schoenfeld [12, Thm. 1] prove

$$\frac{t}{\ln t}\left(1 + \frac{1}{2\ln t}\right) < \pi(t) < \frac{t}{\ln t}\left(1 + \frac{3}{2\ln t}\right) \qquad (t \geq 59),$$

the upper bound holding for all $t > 1$; in particular $|\pi(t) - t/\ln t| \leq \frac{3}{2} t/(\ln t)^2$ for $t \geq 59$. Fix $1 \leq j \leq J := M^{1/4}$ and put $A := M/(j+1)$, $B := M/j$, so that the primes at level j are exactly those in $(A, B]$. There is an absolute M_1 such that for $M \geq M_1$: $A \geq M^{3/4}/2 \geq 59$ and $\ln t \geq \frac{7}{10} \ln M$ for all $t \in [A, B]$; consequently

$$\left| \frac{1}{\ln t} - \frac{1}{\ln M} \right| = \frac{|\ln M - \ln t|}{\ln t \cdot \ln M} \leq \frac{2\ln(j+2)}{(\ln M)^2} \qquad \text{uniformly on } [A, B].$$

Partial summation gives the exact identity $\sum_{p \in (A,B]} p^{-2} = \pi(B)/B^2 - \pi(A)/A^2 + 2\int_A^B \pi(t)\, t^{-3}\, \mathrm{d}t$. Inserting the Rosser–Schoenfeld bound and the displayed estimate for $1/\ln t$ (with $\pi(B)/B^2 = 1/(B \ln B) + O(1/(B(\ln B)^2))$, $1/B = j/M$, and analogously at A, and with $\int_A^B t^{-2}\, \mathrm{d}t = 1/A - 1/B = 1/M$ exactly) yields, with absolute implied constants,

$$\frac{\pi(B)}{B^2} = \frac{j}{M \ln M} + O\left(\frac{(j+1)\ln(j+2)}{M(\ln M)^2}\right), \qquad \frac{\pi(A)}{A^2} = \frac{j+1}{M \ln M} + O\left(\frac{(j+1)\ln(j+2)}{M(\ln M)^2}\right),$$

$$2\int_A^B \frac{\pi(t)}{t^3}\, \mathrm{d}t = \frac{2}{M \ln M} + O\left(\frac{\ln(j+2)}{M(\ln M)^2}\right).$$

Combining the three displays,

$$\sum_{p:\, m_p = j} \frac{1}{p^2} = \frac{1}{M \ln M} + \theta_j, \qquad |\theta_j| \leq \frac{C\,(j+1)\ln(j+2)}{M(\ln M)^2} \qquad (1 \leq j \leq M^{1/4},\ M \geq M_1), \tag{10}$$

with C absolute. (For large j the error term may exceed the main term; only its exponentially weighted sum matters below.)

Step 2: from $1/p^2$ to the true summand. As at the start of Section 3.1, for $j \leq M^{1/4}$,

$$\Big| \sum_{p:\, m_p = j} \ln(1 - 1/p)^2 - \sum_{p:\, m_p = j} \frac{1}{p^2} \Big| \leq C\, \frac{(j+1)^3}{M^3}\, \pi(M/j) \leq \frac{C'(j+1)^2}{M^2 \ln M}.$$

Step 3: exponential weights. Write $w_j := (1 - \frac{j}{M-1})^k (1 - (1 - \frac{j}{M-1})^k)$ and $w_j^\infty := e^{-cj}(1 - e^{-cj})$. For $j \leq M^{1/4}$, expanding $k \ln(1 - j/(M-1)) = -cj + r_j$ with $|r_j| \leq C(c)(j^2 + j)/M \leq 1$ for

$M \geq M_2(c)$ gives $|(1-\frac{j}{M-1})^k - e^{-cj}| \leq 2e^{-cj}|r_j| \leq C(c)\, e^{-cj}(j^2+1)/M$. Since $u \mapsto u(1-u)$ is 1-Lipschitz on $[0,1]$,

$$|w_j - w_j^\infty| \leq C(c)\, e^{-cj}\, \frac{j^2+1}{M} \qquad (1 \leq j \leq M^{1/4},\ M \geq M_2(c)).$$

Step 4: assembly. For the main range $j \leq J$, Steps 1–3 together with $w_j \leq 2e^{-cj}$ (Section 3.1) give

$$\begin{aligned} M \ln M \sum_{j \leq J} w_j \sum_{p\colon m_p = j} \ln(1-1/p)^2 &= \sum_{j \leq J} w_j + O\left(\sum_{j \geq 1} 2e^{-cj}\, \frac{C(j+1)\ln(j+2)}{\ln M} \right) \\ &\qquad + O_c\left(\frac{1}{M} \sum_{j \geq 1} e^{-cj}(j+1)^2 \right) \\ &= \sum_{j \leq J} w_j^\infty + O_c\left(\frac{1}{\ln M} \right) \\ &= B(c) + O\big(e^{-cJ}\big) + O_c\left(\frac{1}{\ln M} \right), \end{aligned}$$

using in the last two equalities Step 3, the convergence of $\sum_j e^{-cj}(j+1)\ln(j+2)$ and $\sum_j e^{-cj} j^2$, and $\sum_{j>J} w_j^\infty \leq \sum_{j>J} e^{-cj} = O(e^{-cJ})$. The tail $j > J$ of V_1 contributes at most $M \ln M \cdot e^{-cM^{1/4}/2} \sum_{p \leq M} \ln(1-1/p)^2 = O_c\big(M \ln M\, e^{-cM^{1/4}/2}\big)$, exactly as in Section 3.1. Therefore $M \ln M \cdot V_1 = B(c) + O_c(1/\ln M)$, which together with (8) completes the proof. $\square$

4 Martingale decomposition

Let $F_k := \ln(\varphi(L_k)/L_k) = \sum_{p \leq M} \ln(1-1/p)\, \mathbf{1}\{p \mid L_k\}$ and $D_i := \mathbb{E}[F_k \mid \mathcal{F}_i] - \mathbb{E}[F_k \mid \mathcal{F}_{i-1}]$ for $1 \leq i \leq k$.

Theorem 13 (Doob decomposition)**.** *$(D_i, \mathcal{F}_i)_{i=1}^k$ is a martingale-difference sequence with $\sum_{i=1}^k D_i = W_k$, and*

$$D_i = \sum_{\substack{p \leq M \\ p \nmid L_{i-1}}} \ln(1-1/p)\, q_p^{k-i}\, \xi_p^{(i)}. \tag{11}$$

Proof. By linearity, $\mathbb{E}[F_k \mid \mathcal{F}_i] = \sum_p \ln(1-1/p)\, \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$. Since $p \mid L_k$ iff $p \mid L_i$ or $p \mid X_j$ for some $j > i$, and $X_{i+1}, \ldots, X_k \perp \mathcal{F}_i$,

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] = \mathbf{1}\{p \mid L_i\} + \mathbf{1}\{p \nmid L_i\}\big(1 - q_p^{k-i}\big).$$

On $\{p \mid L_{i-1}\}$, both $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$ and $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}]$ equal 1, so this prime contributes nothing to D_i. On $\{p \nmid L_{i-1}\}$, case analysis on $\mathbf{1}\{p \mid X_i\} \in \{0,1\}$ gives

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] - \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}] = \begin{cases} q_p^{k-i+1}, & p \mid X_i, \\ -q_p^{k-i}(1-q_p), & p \nmid X_i, \end{cases}$$

which equals $q_p^{k-i}\, \xi_p^{(i)}$ in either case. Summing over p gives (11); martingale-difference and telescoping properties are immediate. $\square$

By orthogonality of martingale increments and Theorem 1, $\sum_{i=1}^k \mathbb{E}[D_i^2] = s_k^2 \sim B(c)/(M \ln M)$.

5 Conditional Lindeberg condition

Let $i_0 := \lceil k/2 \rceil$ and define the event

$$\Omega^* := \bigcap_{\substack{p \le M \\ m_p \ge (\log M)^2}} \{p \mid L_{i_0 - 1}\}.$$

Theorem 14 (Uniform increment bound)**.** *For every fixed $c > 0$ and every $K \ge 0$, $\mathbb{P}((\Omega^*)^c) = o(M^{-K})$ as $M \to \infty$. Moreover, there exists $M_0 = M_0(c)$ such that, for $M \ge M_0$,*

$$\max_{1 \le i \le k} |D_i| \le C(c)\,(\log M)^3/M = o(s_k) \tag{12}$$

on Ω^*. More precisely, for $i < i_0$ the bound holds almost surely, while for $i \ge i_0$ it holds uniformly over the possible value of X_i whenever the history belongs to Ω^*.

Proof. Probability of Ω^*. For a prime p with $m_p \ge (\log M)^2$,

$$\mathbb{P}(p \nmid L_{i_0 - 1}) = q_p^{i_0 - 1} \le \exp\bigl(-\pi_p(i_0 - 1)\bigr) \le \exp\bigl(-(c/2)(\log M)^2(1 + o(1))\bigr).$$

A union bound over at most $\pi(M) \le M/\ln M$ primes gives

$$\mathbb{P}((\Omega^*)^c) \le \frac{M}{\ln M} \exp\bigl(-(c/2)(\log M)^2(1 + o(1))\bigr) = o(M^{-K})$$

for every K.

Case A: $i \ge i_0$, on Ω^*. Decompose $D_i = S_i(X_i) - C_i$, where

$$S_i(x) := \sum_{\substack{p \mid x \\ p \nmid L_{i-1}}} \ln(1 - 1/p)\, q_p^{k-i}, \qquad C_i := \sum_{p \nmid L_{i-1}} \ln(1 - 1/p)\, q_p^{k-i}\, \pi_p.$$

On Ω^*, every contributing prime satisfies $m_p < (\log M)^2$, hence $p > M/(\log M)^2$. For $M \ge M_0(c)$ large enough that $M > (\log M)^4$, any two such primes p_1, p_2 obey

$$p_1 p_2 > \bigl(M/(\log M)^2\bigr)^2 = M^2/(\log M)^4 > M,$$

so p_1 and p_2 cannot both divide any $x \in \{2, \ldots, M\}$. Hence at most one contributing prime divides x, giving

$$|S_i(x)| \le \max_{\substack{p \nmid L_{i-1} \\ m_p < (\log M)^2}} \frac{q_p^{k-i}}{p - 1} \le \frac{(\log M)^2}{M - 1}.$$

For C_i, a prime at level $j \le (\log M)^2$ contributes at most $|\ln(1 - 1/p)|\pi_p \le 1/(p-1) \cdot j/(M-1) \le 2(j+1)/M \cdot (j+1)/M \le 2(j+1)^2/M^2$ (using $q_p^{k-i} \le 1$, the crudest factor since $k - i$ may be as small as zero when $i = k$). Summing using $|\{p : m_p = j\}| \le 1.26\, M/(j \ln(M/j))$ of Lemma 5:

$$|C_i| \le \sum_{j=1}^{(\log M)^2} \frac{1.26\, M}{j \ln(M/j)} \cdot \frac{2(j+1)^2}{M^2} \le \frac{5.04}{M \ln M} \sum_{j=1}^{(\log M)^2} \frac{(j+1)^2}{j} = O\biggl(\frac{(\log M)^3}{M}\biggr),$$

since $\sum_{j=1}^{J} (j+1)^2/j = \sum_j (j + 2 + 1/j) = O(J^2)$ at $J = (\log M)^2$.

Combining, $|D_i| \le |S_i(X_i)| + |C_i| \le C(c)(\log M)^3/M$ uniformly over X_i on Ω^*.

Case B: $i < i_0$, unconditionally. Put $r := k - i$. Since $i < \lceil k/2 \rceil$, one has $r > k/2$; hence, for all sufficiently large M, $r \geq cM/3$. We again write $D_i = S_i(X_i) - C_i$.

For the part of $S_i(x)$ coming from primes $p \geq \sqrt{M}$, at most one such prime can divide a fixed $x \leq M$: two distinct prime divisors in this range would have product $> M$. If that prime has level $m_p = j \leq \sqrt{M}$, then

$$\frac{q_p^r}{p-1} \leq \frac{2(j+1)}{M} \exp\left(-\frac{rj}{M-1}\right) \leq \frac{2(j+1)}{M} e^{-cj/4} = O_c(M^{-1}),$$

uniformly in j. For primes $p < \sqrt{M}$, one has $m_p \geq \sqrt{M} - 1$, so

$$q_p^r \leq \exp\left(-\frac{rm_p}{M-1}\right) \leq e^{-c\sqrt{M}/4}$$

for large M. Consequently their total contribution is at most $e^{-c\sqrt{M}/4} \sum_{p \leq M} (p-1)^{-1} = O(e^{-c\sqrt{M}/4} \log\log M)$. Thus $\sup_x |S_i(x)| = O_c(M^{-1})$.

For the centring term, split by levels. When $j \leq \sqrt{M}$, Lemma 5, $q_p^r \leq e^{-cj/4}$, $\pi_p = j/(M-1)$, and $(p-1)^{-1} \leq 2(j+1)/M$ give

$$\sum_{p:m_p=j} |\ln(1-1/p)|\, \pi_p q_p^r \leq \frac{C}{M \ln M}(j+1)e^{-cj/4}.$$

Summing over j yields $O_c((M \ln M)^{-1})$. The levels $j > \sqrt{M}$ contribute $O(e^{-c\sqrt{M}/4} \log\log M)$. Therefore $|C_i| = O_c((M \ln M)^{-1})$ and $|D_i| = O_c(M^{-1})$ uniformly over all histories and all values of X_i.

Comparison to s_k. $(\log M)^3/M \cdot (M \ln M)^{1/2} = (\log M)^{7/2}/\sqrt{M} \to 0$, establishing (12). □

Corollary 15 (Conditional Lindeberg)**.** *For every $\varepsilon > 0$,*

$$\frac{1}{s_k^2} \sum_{i=1}^{k} \mathbb{E}\left[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}\right] \xrightarrow{P} 0.$$

Proof. By Remark 11, $s_k \asymp (M \ln M)^{-1/2}$, and hence $(\log M)^3/M = o(s_k)$. For M sufficiently large, Theorem 14 implies that every summand in the conditional Lindeberg expression vanishes on Ω^*: for $i < i_0$ this holds unconditionally, and for $i \geq i_0$ it holds uniformly over X_i conditional on any history in Ω^*. Therefore the entire nonnegative conditional Lindeberg sum is zero on Ω^*. Consequently, for every $\delta > 0$,

$$\mathbb{P}\left(\frac{1}{s_k^2} \sum_{i=1}^{k} \mathbb{E}\left[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}\right] > \delta\right) \leq \mathbb{P}((\Omega^*)^c) \longrightarrow 0.$$

□

6 Conditional-variance concentration

Let $U_i := \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]$. Expanding via (11),

$$\sum_{i=1}^{k} U_i = A + 2B$$

where

$$A := \sum_{p \le M} c_{pp}\, R^{(k)}_{pp}, \qquad B := \sum_{p<q\le M} c_{pq}\, R^{(k)}_{pq},$$

with $c_{pq} := \ln(1-1/p)\ln(1-1/q)\,\kappa_{pq}$, $\kappa_{pp} := \pi_p q_p$, $\kappa_{pq} := \beta_{\{p,q\}} - q_p q_q$ for $p \ne q$, and

$$R^{(k)}_{pp} := \sum_{i=1}^{k} \mathbf{1}\{T_p \ge i\}\, q_p^{2(k-i)}, \qquad R^{(k)}_{pq} := \sum_{i=1}^{k} \mathbf{1}\{T_p \ge i,\, T_q \ge i\}\,(q_p q_q)^{k-i}.$$

By Proposition 6, $\kappa_{pq} = -\pi_p \pi_q$ whenever $pq > M$.

The main estimates of this section are the following.

Lemma 16 (Per-prime variance)**.** *For every $M \ge 51$, every prime $p \le M$, and every $k \ge 1$,*

$$\operatorname{Var}(R^{(k)}_{pp}) \le \frac{3\, q_p^k}{\pi_p^2}. \tag{13}$$

Lemma 17 (Cross-prime covariance)**.** *There exists a constant $C_0(c) > 0$ (depending only on c) such that, for distinct primes $p, q \le M$ with $pq > M$ and $\pi_p + \pi_q \le 1/3$, in the joint regime $k = \lfloor cM \rfloor$,*

$$\left|\operatorname{Cov}(R^{(k)}_{pp}, R^{(k)}_{qq})\right| \le C_0(c)\, k\, q_p^k q_q^k. \tag{14}$$

An admissible value is $C_0(c) = 57 + 54/c$; this is what the proof chain yields directly, though numerical evidence suggests that it is not sharp. Only the shape of the bound — linear in k, multiplicative in $q_p^k q_q^k$ — is used in Theorem 19 below.

6.1 Proof of Lemma 16

Set $Y_p := q_p^{-2(T_p \wedge k)}$. Then $R^{(k)}_{pp} = q_p^{2k}(Y_p - 1)/(1 - q_p^2)$, and

$$\operatorname{Var}(R^{(k)}_{pp}) = \frac{q_p^{4k}\, \operatorname{Var}(Y_p)}{(1-q_p^2)^2} \le \frac{q_p^{4k}\, \mathbb{E}[Y_p^2]}{(1-q_p^2)^2}.$$

Compute

$$\mathbb{E}[Y_p^2] = \sum_{m=1}^{k} q_p^{-4m}\, \pi_p q_p^{m-1} + q_p^{-3k} = \frac{\pi_p}{q_p} \cdot \frac{q_p^{-3k} - 1}{1 - q_p^3} + q_p^{-3k}.$$

Since $1 - q_p^3 = \pi_p(1 + q_p + q_p^2)$, the first term simplifies to $(q_p^{-3k} - 1)/(q_p(1 + q_p + q_p^2)) \le q_p^{-3k}/(q_p(1 + q_p + q_p^2))$. Every prime $p \ge 2$ has $m_p \le M/2$, so $\pi_p \le (M/2)/(M-1) \le 51/100$ for $M \ge 51$, and hence $q_p \ge 49/100$ for *every* prime $p \le M$. Consequently $q_p(1 + q_p + q_p^2) \ge 0.49\,(1 + 0.49 + 0.49^2) > 4/5$, so

$$\mathbb{E}[Y_p^2] \le q_p^{-3k}\big(5/4 + 1\big) = (9/4)\, q_p^{-3k} \le 3\, q_p^{-3k}.$$

Using $(1 - q_p^2)^2 = \pi_p^2(1 + q_p)^2 \ge \pi_p^2$:

$$\operatorname{Var}(R^{(k)}_{pp}) \le \frac{q_p^{4k} \cdot 3 q_p^{-3k}}{\pi_p^2} = \frac{3\, q_p^k}{\pi_p^2}.$$

6.2 Proof of Lemma 17

Write $\mathbb{P}_{\text{dep}}$ for the joint law of (T_p, T_q) under the true distribution and $\mathbb{P}_{\text{ind}}$ for the product of the (common) marginals. By Proposition 6, at each step three outcomes are possible with probabilities β, π_p, π_q (where $\beta := \beta_{\{p,q\}} = 1 - \pi_p - \pi_q$), giving, for $1 \le i < j \le k$,

$$\mathbb{P}_{\text{dep}}(T_p = i, T_q = j) = \beta^{i-1}\,\pi_p\, q_q^{j-i-1}\,\pi_q,$$

and the analogous formula for $j < i$, with $\mathbb{P}_{\text{dep}}(T_p = T_q < \infty) = 0$ (disjointness). Under $\mathbb{P}_{\text{dep}}$, the marginal law of T_p is geometric with parameter π_p (at each step the event $\{p \mid X\}$ has probability π_p), and similarly for T_q. The product law with identical marginals is then

$$\mathbb{P}_{\text{ind}}(T_p = i, T_q = j) = \pi_p q_p^{i-1} \cdot \pi_q q_q^{j-1} = \pi_p \pi_q (q_p q_q)^{i-1} q_q^{j-i} \qquad (i < j),$$

and analogously for $j < i$ and for the diagonal $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p\pi_q(q_pq_q)^{m-1}$. Set $\delta := \pi_p\pi_q/(q_pq_q)$, so $\beta/(q_pq_q) = 1 - \delta$. Under $\pi_p + \pi_q \le 1/3$, $q_pq_q \ge (1-\pi_p)(1-\pi_q) \ge 1 - (\pi_p + \pi_q) \ge 2/3$, whence $\delta \le (3/2)\pi_p\pi_q$ and $1/q_p, 1/q_q \le 3/2$. Since marginals agree, $\mathrm{Cov}_{\text{dep}}(Y_p, Y_q) = \mathbb{E}_{\text{dep}}[Y_pY_q] - \mathbb{E}_{\text{ind}}[Y_pY_q]$, where $Y_p := q_p^{-2(T_p \wedge k)}$, $Y_q := q_q^{-2(T_q \wedge k)}$.

Partition $\{(i,j) : 1 \le i, j \le k+1\}$ (with $T > k$ encoded as $T = k+1$) into the regions $i < j$, $i > j$, $i = j \le k$, and $i = j = k+1$.

Region $i < j$. The density ratio is

$$\frac{\mathbb{P}_{\text{dep}}(T_p = i, T_q = j)}{\mathbb{P}_{\text{ind}}(T_p = i, T_q = j)} = \frac{(1-\delta)^{i-1}}{q_q}.$$

Write $(1-\delta)^{i-1}/q_q - 1 = [(1-\delta)^{i-1} - 1]/q_q + \pi_q/q_q$. Using $|(1-\delta)^{i-1} - 1| \le (i-1)\delta$,

$$\begin{aligned}&|\mathbb{E}_{\text{dep}}[Y_pY_q; i<j] - \mathbb{E}_{\text{ind}}[Y_pY_q; i<j]| \\ &\qquad \le (\delta/q_q)\,\mathbb{E}_{\text{ind}}[(T_p - 1)Y_pY_q; i<j] + (\pi_q/q_q)\,\mathbb{E}_{\text{ind}}[Y_pY_q; i<j].\end{aligned}$$

Under $\mathbb{P}_{\text{ind}}$, independence gives $\mathbb{E}[T_pY_pY_q; i<j] \le \mathbb{E}[T_pY_p]\,\mathbb{E}[Y_q]$. For $r = q_p^{-1} > 1$: $\sum_{m=1}^k mr^m \le kr^{k+1}/(r-1)$ (telescoping identity $(r-1)\sum_m mr^m = kr^{k+1} - \sum_m r^m$, dropping the positive term). Hence

$$\mathbb{E}[T_pY_p] = \frac{\pi_p}{q_p}\sum_{m=1}^{k} mq_p^{-m} + (k+1)q_p^{-k} \le \frac{\pi_p}{q_p}\cdot\frac{kq_p^{-k-1}}{\pi_p} + 2kq_p^{-k} = kq_p^{-k-1} + 2kq_p^{-k} \le 4kq_p^{-k},$$

for $q_p \ge 1/2$. Analogously $\mathbb{E}[Y_q] \le 3q_q^{-k}$ (from $\mathbb{E}[Y_q] = (1 + 1/q_q)q_q^{-k} - 1/q_q \le 3q_q^{-k}$) and $\mathbb{E}[Y_p] \le 3q_p^{-k}$. Thus

$$\mathbb{E}_{\text{ind}}[T_pY_pY_q; i<j] \le 12kq_p^{-k}q_q^{-k}, \qquad \mathbb{E}_{\text{ind}}[Y_pY_q; i<j] \le 9\,q_p^{-k}q_q^{-k}.$$

Using $\delta/q_q \le (3/2)(3/2)\pi_p\pi_q = (9/4)\pi_p\pi_q$ and $\pi_q/q_q \le (3/2)\pi_q$:

$$|\mathbb{E}_{\text{dep}}[Y_pY_q; i<j] - \mathbb{E}_{\text{ind}}[Y_pY_q; i<j]| \le 27\,k\pi_p\pi_q\,q_p^{-k}q_q^{-k} + \tfrac{27}{2}\,\pi_q\,q_p^{-k}q_q^{-k}. \tag{15}$$

Region $i > j$. By symmetry (swap roles of p and q):

$$|\mathbb{E}_{\text{dep}}[Y_pY_q; i>j] - \mathbb{E}_{\text{ind}}[Y_pY_q; i>j]| \le 27\,k\pi_p\pi_q\,q_p^{-k}q_q^{-k} + \tfrac{27}{2}\,\pi_p\,q_p^{-k}q_q^{-k}. \tag{16}$$

Region $i = j \le k$. Here $\mathbb{P}_{\text{dep}} = 0$ and $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p \pi_q (q_p q_q)^{m-1}$, contributing

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \le k] = \pi_p \pi_q \sum_{m=1}^{k} (q_p q_q)^{m-1} (q_p q_q)^{-2m} = \pi_p \pi_q (q_p q_q)^{-1} \sum_{m=1}^{k} (q_p q_q)^{-m},$$

and using $\sum_{m=1}^{k} r^m \le k r^k$ for $r = (q_p q_q)^{-1} \ge 1$,

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \le k] \le \pi_p \pi_q k (q_p q_q)^{-k-1} \le \tfrac{3}{2} k \pi_p \pi_q \, q_p^{-k} q_q^{-k}. \tag{17}$$

Region $i = j = k+1$. Here $\mathbb{P}_{\text{dep}}(T_p > k, T_q > k) = \beta^k$ and $\mathbb{P}_{\text{ind}}(T_p > k, T_q > k) = (q_p q_q)^k$, with $Y_p Y_q = q_p^{-2k} q_q^{-2k}$:

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| = (q_p q_q)^{-2k} |\beta^k - (q_p q_q)^k|.$$

Using $|\beta^k - (q_p q_q)^k| = (q_p q_q)^k |1 - (1-\delta)^k| \le (q_p q_q)^k k\delta$ and $\delta \le (3/2)\pi_p \pi_q$,

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| \le \tfrac{3}{2} k \pi_p \pi_q \, (q_p q_q)^{-k} \le \tfrac{3}{2} k \pi_p \pi_q \, q_p^{-k} q_q^{-k}. \tag{18}$$

Assembly. Combining (15)–(18),

$$|\operatorname{Cov}(Y_p, Y_q)| \le 57 \, k \pi_p \pi_q \, q_p^{-k} q_q^{-k} + \tfrac{27}{2} (\pi_p + \pi_q) \, q_p^{-k} q_q^{-k}. \tag{19}$$

(The coefficient $57 = 27 + 27 + 3/2 + 3/2$ dominates the individual contributions.) Using the linear relation $R_{pp}^{(k)} = q_p^{2k}(Y_p - 1)/(1 - q_p^2)$ from the proof of Lemma 16 (analogously for q) and $(1 - q_p^2)(1 - q_q^2) = \pi_p \pi_q (1 + q_p)(1 + q_q) \ge \pi_p \pi_q$:

$$|\operatorname{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| = \frac{(q_p q_q)^{2k} \, |\operatorname{Cov}(Y_p, Y_q)|}{(1 - q_p^2)(1 - q_q^2)} \le \frac{|\operatorname{Cov}(Y_p, Y_q)| \, (q_p q_q)^{2k}}{\pi_p \pi_q}.$$

Substituting (19):

$$|\operatorname{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \le \left[57 \, k + \tfrac{27}{2} \left(\tfrac{1}{\pi_q} + \tfrac{1}{\pi_p} \right) \right] q_p^k q_q^k.$$

Since $\pi_p \ge 1/(M-1)$ trivially and $k = \lfloor cM \rfloor \ge cM - 1$, we have $M \le (k+1)/c$, so $1/\pi_p \le M - 1 \le (k+1)/c$, and likewise for q. Therefore $1/\pi_p + 1/\pi_q \le 2(k+1)/c$, and using $k + 1 \le 2k$ for $k \ge 1$,

$$|\operatorname{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \le [57 + 54/c] \, k \, q_p^k q_q^k.$$

Thus $C_0(c) := 57 + 54/c$ suffices for all $k \ge 1$. □

6.3 Level-summation bound and assembly

Lemma 18 (Level-summation upper bound)**.** *Let $f \colon \mathbb{N}_{\ge 1} \to \mathbb{R}_{\ge 0}$ satisfy $f(j) \le A(1+j)^r$ for $A \ge 0$, $r \ge 0$. For each $c > 0$ there exist $M_0(c, r)$ and $C(c, r)$ such that for $M \ge M_0(c, r)$,*

$$L_M(f) := \sum_{p \le M} \frac{f(m_p) \, q_p^{\lfloor cM \rfloor}}{(p-1)^4} \le \frac{C(c, r) \, A}{M^3 \ln M}. \tag{20}$$

Proof. Partitioning by level and bounding $1/(p-1)^4 \le 16(j+1)^4/M^4$ for $M \ge 2(j+1)$, together with Lemma 5:

$$S_j(M) := \sum_{p:\, m_p=j} \frac{1}{(p-1)^4} \le \frac{20.16(j+1)^4}{j\ln(M/j)\, M^3}.$$

For $j \le M^{1/2}$, $\ln(M/j) \ge (\ln M)/2$, so $S_j(M) \le 40.32(j+1)^4/(j\ln M \cdot M^3)$.

For the exponential factor, $\ln(1-x) \le -x$ (concavity) gives $\ln(1-j/(M-1))^{\lfloor cM\rfloor} \le -cj + O(j/M)$ uniformly for $j \le M^{1/2}$, so $(1-j/(M-1))^{\lfloor cM\rfloor} \le 2e^{-cj}$ for $M \ge M_1(c)$.

Main range $j \le M^{1/2}$:

$$\sum_{j=1}^{\lfloor M^{1/2}\rfloor} f(j)\cdot(1-j/(M-1))^{\lfloor cM\rfloor}\, S_j(M) \le \frac{80.64}{M^3\ln M}\sum_{j\ge 1}\frac{A(1+j)^{r+4}\, e^{-cj}}{j},$$

and the series converges to a finite $\Sigma(c,r) < \infty$.

Tail $j > M^{1/2}$: $q_p^k \le e^{-c\sqrt{M}/2}$ times $\sum_p 1/(p-1)^4 = O(1)$ times $\sum_{j>M^{1/2}}^{M} A(1+j)^r = O(AM^{r+1})$, giving super-polynomially small total. Absorbing into the main-range bound yields (20) with $C(c,r) = 81\,\Sigma(c,r)$. □

Theorem 19 (Conditional-variance concentration)**.** *As $M\to\infty$ in the joint regime,*

$$\mathrm{Var}\left(\sum_{i=1}^{k} U_i\right) = O_c\left(\frac{1}{M^2(\ln M)^4}\right) = o(s_k^4).$$

Consequently $\sum_i U_i/s_k^2 \xrightarrow{P} 1$.

Proof. We have

$$\mathrm{Var}\left(\sum_i U_i\right) = \mathrm{Var}(A) + 4\,\mathrm{Cov}(A,B) + 4\,\mathrm{Var}(B).$$

Set $J := M^{1/4}$. Terms involving a prime with $m_p > J$ will be called tail terms. Since

$$q_p^k \le \exp\left(-\frac{km_p}{M-1}\right) \le e^{-cJ/2}$$

for such primes and all sufficiently large M, every tail estimate below is super-polynomially small; polynomial factors in M and powers of $\log\log M$ are therefore harmless.

Bound on $\mathrm{Var}(A)$. By Lemma 16, which applies to every prime $p \le M$ once $M \ge 51$,

$$c_{pp}^2\,\mathrm{Var}(R_{pp}^{(k)}) \le \frac{\pi_p^2}{(p-1)^4}\frac{3q_p^k}{\pi_p^2} = \frac{3q_p^k}{(p-1)^4}.$$

Lemma 18 therefore gives

$$\sum_p c_{pp}^2\,\mathrm{Var}(R_{pp}^{(k)}) = O_c\left(\frac{1}{M^3\ln M}\right).$$

For the covariance terms with $m_p, m_q \le J$, we have $pq > M$ and $\pi_p + \pi_q \le 1/3$ for all sufficiently large M, so Lemma 17 applies. Hence

$$\sum_{\substack{p\ne q\\ m_p,m_q\le J}} |c_{pp}c_{qq}\,\mathrm{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \le C_0(c)k\left(\sum_p \frac{\pi_p q_p^k}{(p-1)^2}\right)^2 = O_c\left(\frac{1}{M^3(\ln M)^2}\right),$$

To justify the last estimate explicitly, set

$$S_M := \sum_{p \le M} \frac{\pi_p q_p^k}{(p-1)^2}.$$

At level $m_p = j \le M^{1/2}$, use $\pi_p = j/(M-1)$, $(p-1)^{-2} \le 4(j+1)^2/M^2$, the count in Lemma 5, and $q_p^k \le 2e^{-cj}$ to obtain

$$S_M \le \frac{C}{M^2 \ln M} \sum_{j \ge 1} (j+1)^2 e^{-cj} + O(e^{-c\sqrt{M}/2}) = O_c\left(\frac{1}{M^2 \ln M}\right).$$

Since $k = O_c(M)$, this gives $kS_M^2 = O_c(M^{-3}(\ln M)^{-2})$ as claimed. If at least one of m_p, m_q exceeds J, Cauchy–Schwarz and Lemma 16 give

$$|c_{pp}|\sqrt{\mathrm{Var}(R_{pp}^{(k)})} \le \frac{\sqrt{3}\, q_p^{k/2}}{(p-1)^2},$$

and the resulting double sum is $O_c(e^{-cJ/4})$. Thus

$$\mathrm{Var}(A) = O_c\left(\frac{1}{M^3 \ln M}\right). \tag{21}$$

A single-pair bound for $R_{pq}^{(k)}$. First suppose $m_p, m_q \le J$. Then $pq > M$, so the events $\{p \mid X\}$ and $\{q \mid X\}$ are disjoint. Let $\beta = 1 - \pi_p - \pi_q$, $a = q_p q_q$, and $N = (T_p \wedge T_q) \wedge k$. Since

$$R_{pq}^{(k)} = \frac{a^k(a^{-N} - 1)}{1 - a},$$

we can bound its second moment directly. Put $u := \pi_p + \pi_q$ and $d := \pi_p \pi_q$, so that $\beta = 1 - u$ and $a = 1 - u + d$. Because $u \le 1/3$ and $d \le u^2/4$, one has $a^2 \le \beta$ and $1 - a = u - d \ge u/2$. Therefore

$$\begin{aligned} a^{2k}\mathbb{E}[a^{-2N}] &= u \sum_{m=1}^{k} a^{2(k-m)} \beta^{m-1} + \beta^k \\ &\le ku\beta^{k-1} + \beta^k \le C(c)ku\beta^{k-1}, \end{aligned}$$

where the last step uses $ku \ge c/2$ for all sufficiently large M. Since $(a^{-N} - 1)^2 \le a^{-2N}$,

$$\mathrm{Var}(R_{pq}^{(k)}) \le \mathbb{E}[(R_{pq}^{(k)})^2] \le \frac{C(c)k\,\beta^{k-1}}{\pi_p + \pi_q}. \tag{22}$$

At a level pair (j_1, j_2) with $j_1, j_2 \le J$ one has $\pi_p + \pi_q = (j_1 + j_2)/(M-1)$, and, since $1 - x \le e^{-x}$ and $(k-1)/(M-1) = c + O(1/M)$,

$$\beta^{k-1} \le \exp\left(-\frac{(k-1)(j_1 + j_2)}{M-1}\right) \le 2\, e^{-c(j_1+j_2)}$$

for $M \ge M_0(c)$, uniformly over $j_1, j_2 \le J$ (as in (6)). Together with $|c_{pq}| \le \pi_p \pi_q/((p-1)(q-1)) \le 16(1+j_1)^2(1+j_2)^2/M^4$, the bound (22) gives

$$|c_{pq}|\sqrt{\mathrm{Var}(R_{pq}^{(k)})} \le \frac{C(c)}{M^3} \frac{(1+j_1)^2(1+j_2)^2}{\sqrt{j_1 + j_2}} e^{-c(j_1+j_2)/2}. \tag{23}$$

Bound on $\mathrm{Var}(B)$. Write $B = B_{\mathrm{main}} + B_{\mathrm{tail}}$, where B_{main} contains pairs with $m_p, m_q \le J$. Minkowski's inequality in L^2 gives

$$\sqrt{\mathrm{Var}(B_{\mathrm{main}})} \le \sum_{\substack{p<q \\ m_p,m_q \le J}} |c_{pq}| \sqrt{\mathrm{Var}(R_{pq}^{(k)})}.$$

By Lemma 5, the number of primes at level $j \le J$ is at most $2.52M/(j \ln M)$. Combining this count with (23),

$$\sqrt{\mathrm{Var}(B_{\mathrm{main}})} \le \frac{C(c)}{M(\ln M)^2} \sum_{j_1,j_2 \ge 1} \frac{(1+j_1)^2(1+j_2)^2}{j_1 j_2 \sqrt{j_1+j_2}} e^{-c(j_1+j_2)/2} = O_c\left(\frac{1}{M(\ln M)^2}\right),$$

because the double series converges.

It remains to control B_{tail}, which also includes every pair with $pq \le M$. For arbitrary distinct p, q, let

$$\beta_{pq} := \mathbb{P}(p \nmid X, q \nmid X), \qquad a := q_p q_q, \qquad N := (T_p \wedge T_q) \wedge k.$$

If p is chosen so that $m_p = \max(m_p, m_q)$, then $a \le q_p$ and $\beta_{pq} \le q_p$. Directly from the geometric law of $T_p \wedge T_q$,

$$a^{2k}\mathbb{E}[a^{-2N}] = (1-\beta_{pq}) \sum_{m=1}^{k} a^{2(k-m)} \beta_{pq}^{m-1} + \beta_{pq}^k \le (k+1) q_p^{k-1}.$$

Since $1 - a \ge \pi_p$,

$$\mathrm{Var}(R_{pq}^{(k)}) \le \mathbb{E}[(R_{pq}^{(k)})^2] \le \frac{(k+1)q_p^{k-1}}{\pi_p^2}. \tag{24}$$

For every tail pair, $m_p > J$ after relabelling. Using the crude bounds $|\kappa_{pq}| \le 1$, $|\ln(1-1/r)| \le 1/(r-1)$, $1/\pi_p \le M$, and $\sum_{\substack{r \le M \\ r \text{ prime}}} (r-1)^{-1} = O(\log\log M)$, Minkowski's inequality and (24) give

$$\sqrt{\mathrm{Var}(B_{\mathrm{tail}})} \le CM^{3/2}(\log\log M)^2 e^{-cJ/4} = o(M^{-K})$$

for every fixed K. Therefore

$$\mathrm{Var}(B) = O_c\left(\frac{1}{M^2(\ln M)^4}\right). \tag{25}$$

This argument controls all covariances among distinct prime-pair summands through Minkowski's inequality; no unproved orthogonality is used.

Conclusion. By Cauchy–Schwarz and (21)–(25),

$$|\mathrm{Cov}(A,B)| \le \sqrt{\mathrm{Var}(A)\,\mathrm{Var}(B)} = O_c\left(\frac{1}{M^{5/2}(\ln M)^{5/2}}\right) = o\left(\frac{1}{M^2(\ln M)^2}\right).$$

Thus

$$\mathrm{Var}\left(\sum_i U_i\right) = O_c\left(\frac{1}{M^2(\ln M)^4}\right) = o(s_k^4),$$

because $s_k^4 \asymp M^{-2}(\ln M)^{-2}$ by Theorem 1. Finally, $\mathbb{E}[\sum_i U_i] = \sum_i \mathbb{E}[D_i^2] = s_k^2$, so Chebyshev's inequality gives $\sum_i U_i / s_k^2 \xrightarrow{P} 1$. $\square$

Lemma 20 (Linear-scale transfer)**.** *Let $Y_k := \varphi(L_k)/L_k$ and $\mu_k := \mathbb{E} \ln Y_k$. In the joint regime,*

$$\frac{e^{\mu_k}}{P_M} \longrightarrow 1, \qquad \frac{Y_k - \rho_k(M)}{P_M s_k} - \frac{W_k(M)}{s_k} \xrightarrow{P} 0.$$

Consequently the log-scale central limit theorem implies the linear-scale central limit theorem stated in Theorem 2.

Proof. Writing $a_p := \ln(1 - 1/p) < 0$,

$$\mu_k - \ln P_M = -\sum_{p \le M} a_p q_p^k = \sum_{p \le M} |a_p| q_p^k.$$

Partitioning by $m_p = j$, using $|a_p| \le 1/(p-1)$ and Lemma 5 for $j \le M^{1/2}$, gives

$$\sum_{p \le M} |a_p| q_p^k \le \frac{C(c)}{\ln M} \sum_{j \ge 1} e^{-cj} + o(1/\ln M) = O_c(1/\ln M).$$

The range $j > M^{1/2}$ is super-polynomially small. Hence $e^{\mu_k}/P_M = 1 + O_c(1/\ln M)$. By Mertens' product theorem [14], $P_M \asymp 1/\ln M$, so also $e^{-\mu_k} = O(\ln M)$.

Since $Y_k = e^{\mu_k + W_k} \le 1$, we have $e^{W_k} \le e^{-\mu_k}$. For $R_k := e^{W_k} - 1 - W_k$, Taylor's inequality gives

$$0 \le R_k \le \tfrac{1}{2} e^{(W_k)_+} W_k^2 \le \tfrac{1}{2} e^{-\mu_k} W_k^2.$$

Therefore

$$\frac{\mathbb{E}|R_k|}{s_k} \le \frac{1}{2} e^{-\mu_k} s_k = O\left(\sqrt{\frac{\ln M}{M}}\right) \longrightarrow 0,$$

where Theorem 1 was used. Since $\rho_k(M) = e^{\mu_k} \mathbb{E}[e^{W_k}]$,

$$\frac{Y_k - \rho_k(M)}{P_M s_k} = \frac{e^{\mu_k}}{P_M} \left(\frac{W_k}{s_k} + \frac{R_k - \mathbb{E} R_k}{s_k} \right).$$

The second term in parentheses tends to zero in L^1, while W_k/s_k is tight because it has variance one. Together with $e^{\mu_k}/P_M \to 1$, this proves the claim. The same estimate also yields the Jensen-gap bound $0 \le \rho_k(M) - e^{\mu_k} \le s_k^2/2 = o(P_M s_k)$. □

7 Proof of Theorem 2

For each M, define the row of the triangular array by $Z_{M,i} := D_i/s_k$, $1 \le i \le k = \lfloor cM \rfloor$, with row filtration $\mathcal{F}_{M,i} := \mathcal{F}_i$. Theorem 13 gives $\mathbb{E}[Z_{M,i} \mid \mathcal{F}_{M,i-1}] = 0$ and $\sum_i Z_{M,i} = W_k/s_k$. We use the Hall–Heyde martingale central limit theorem [5, Thm. 3.2] in the following standard form: it is enough that, for every $\varepsilon > 0$,

$$\sum_{i=1}^{k} \mathbb{E}\left[Z_{M,i}^2 \mathbf{1}\{|Z_{M,i}| > \varepsilon\} \mid \mathcal{F}_{M,i-1}\right] \xrightarrow{P} 0,$$

and that

$$\sum_{i=1}^{k} \mathbb{E}[Z_{M,i}^2 \mid \mathcal{F}_{M,i-1}] \xrightarrow{P} 1.$$

Corollary 15 verifies the first condition and Theorem 19 verifies the second. Hence $W_k/s_k \xrightarrow{d} \mathcal{N}(0,1)$. Theorem 1 and Slutsky's theorem give $\sqrt{M \ln M}\, W_k/\sqrt{B(c)} \xrightarrow{d} \mathcal{N}(0,1)$. Lemma 20 gives the corresponding linear-scale limit for $\varphi(L_k)/L_k - \rho_k(M)$.

8 A Berry–Esseen rate: proof of Theorem 3

We use the martingale Berry–Esseen inequality of Heyde and Brown [6] (see also [5, Thm. 3.7]): if $(Z_i, \mathcal{F}_i)_{i=1}^n$ is a square-integrable martingale-difference sequence with $S_n := \sum_{i=1}^n Z_i$, $\mathbb{E}S_n^2 = 1$, and $V_n^2 := \sum_{i=1}^n \mathbb{E}[Z_i^2 \mid \mathcal{F}_{i-1}]$, then for each $\delta \in (0,1]$ there is a constant C_δ, depending only on δ, such that

$$\sup_{x\in\mathbb{R}} \left|\mathbb{P}(S_n \le x) - \Phi(x)\right| \;\le\; C_\delta \left(\sum_{i=1}^n \mathbb{E}|Z_i|^{2+2\delta} + \mathbb{E}\left|V_n^2 - 1\right|^{1+\delta} \right)^{1/(3+2\delta)}. \tag{26}$$

We apply (26) with $\delta = 1$ to $Z_i := D_i/s_k$, $n = k$, for which $\mathbb{E}S_k^2 = 1$ and $\mathbb{E}V_k^2 = 1$ exactly, by orthogonality of the martingale increments (Section 4). The two inputs are supplied by the following lemma and by Theorem 19.

Lemma 21 (Fourth-moment bound)**.** *For each fixed $c > 0$, in the joint regime,*

$$\frac{1}{s_k^4} \sum_{i=1}^k \mathbb{E}D_i^4 \;=\; O_c\!\left(\frac{(\ln M)^7}{M} \right).$$

Proof. Let $b_M := C(c)(\log M)^3/M$ be the bound of Theorem 14, so that $|D_i| \le b_M$ pointwise on Ω^* for every $1 \le i \le k$ (for $i < i_0$ the stronger unconditional bound $|D_i| \le C(c)/M$ holds). Hence

$$\mathbb{E}\left[D_i^4 \mathbf{1}_{\Omega^*}\right] \le b_M^2\, \mathbb{E}\left[D_i^2\right],$$

while on the complement the crude bound $|D_i| \le 2\Lambda(M) = O(\log\log M)$ together with $\mathbb{P}((\Omega^*)^c) = o(M^{-12})$ (Theorem 14) gives $\mathbb{E}[D_i^4 \mathbf{1}_{(\Omega^*)^c}] = o(M^{-11})$. Summing over $i \le k$ and using $\sum_i \mathbb{E}D_i^2 = s_k^2$,

$$\sum_{i=1}^k \mathbb{E}D_i^4 \le b_M^2\, s_k^2 + o(M^{-10}).$$

Since $s_k^{-2} = O_c(M \ln M)$ by Remark 11,

$$\frac{1}{s_k^4} \sum_{i=1}^k \mathbb{E}D_i^4 \le \frac{b_M^2}{s_k^2} + o(M^{-10})\, s_k^{-4} = O_c\!\left(\frac{(\log M)^6}{M^2} \cdot M \ln M \right) + o(M^{-7}) = O_c\!\left(\frac{(\ln M)^7}{M} \right). \qquad \square$$

Proof of Theorem 3. Step 1: the self-normalised bound. Apply (26) with $\delta = 1$ and $Z_i = D_i/s_k$. The first term is $O_c((\ln M)^7/M)$ by Lemma 21. For the second, since $\mathbb{E}V_k^2 = 1$,

$$\mathbb{E}\left(V_k^2 - 1\right)^2 = \frac{1}{s_k^4} \operatorname{Var}\!\left(\sum_{i=1}^k U_i \right) = O_c\!\left(M^2 (\ln M)^2 \cdot \frac{1}{M^2 (\ln M)^4} \right) = O_c\!\left(\frac{1}{(\ln M)^2} \right),$$

by Theorem 19 and Remark 11. Hence

$$\sup_x \left| \mathbb{P}\left(W_k(M)/s_k \le x\right) - \Phi(x) \right| \le C_1 \Big(O_c\big((\ln M)^7/M\big) + O_c\big((\ln M)^{-2}\big) \Big)^{1/5} = O_c\big((\ln M)^{-2/5}\big). \tag{27}$$

Choosing $\delta < 1$ in (26) only worsens the exponent $(1+\delta)/(3+2\delta)$ of the dominant second term, so $\delta = 1$ is optimal here.

Step 2: two elementary stability estimates. We record: (a) for any random variables Z, Δ and any $\varepsilon > 0$,

$$\sup_x \left|\mathbb{P}(Z + \Delta \le x) - \Phi(x)\right| \le \sup_x \left|\mathbb{P}(Z \le x) - \Phi(x)\right| + \frac{\varepsilon}{\sqrt{2\pi}} + \mathbb{P}(|\Delta| > \varepsilon); \tag{28}$$

(b) for deterministic a with $|a-1| \le 1/2$ and any random variable Z',

$$\sup_x \big|\mathbb{P}(aZ' \le x) - \Phi(x)\big| \le \sup_x \big|\mathbb{P}(Z' \le x) - \Phi(x)\big| + 2\,|a-1|. \tag{29}$$

Both are standard. For (a), $\{Z+\Delta \le x\} \subseteq \{Z \le x+\varepsilon\} \cup \{|\Delta| > \varepsilon\}$ and $\{Z \le x-\varepsilon\} \subseteq \{Z+\Delta \le x\} \cup \{|\Delta| > \varepsilon\}$, combined with $0 \le \Phi(x \pm \varepsilon) - \Phi(x) \lessgtr \pm\varepsilon/\sqrt{2\pi}$. For (b), $\mathbb{P}(aZ' \le x) = \mathbb{P}(Z' \le x/a)$, and $\sup_x |\Phi(x/a) - \Phi(x)| \le \frac{|1-a|}{\min(a,1)} \sup_u |u|\phi_{\mathcal{N}}(u) \le 2\,|a-1|\,\phi_{\mathcal{N}}(1) \le |a-1|$, where $\phi_{\mathcal{N}}$ is the standard normal density.

Step 3: renormalisation and the linear scale. Let $\sigma_M := \sqrt{B(c)/(M\ln M)}$ and $r_M := s_k/\sigma_M$. Proposition 12 gives $r_M^2 = 1 + O_c(1/\ln M)$, hence $|r_M - 1| = O_c(1/\ln M)$ for $M \ge M_0(c)$. Since $\sqrt{M\ln M}\,W_k/\sqrt{B(c)} = r_M \cdot (W_k/s_k)$, the estimates (29) and (27) give the second form of the theorem, with an additional error $O_c(1/\ln M)$ absorbed into $O_c((\ln M)^{-2/5})$.

For the linear scale, write, as in the proof of Lemma 20,

$$\frac{Y_k - \rho_k(M)}{P_M s_k} = a_M\left(\frac{W_k}{s_k} + \Delta\right), \qquad a_M := \frac{e^{\mu_k}}{P_M}, \quad \Delta := \frac{R_k - \mathbb{E}R_k}{s_k},$$

where, by that proof, $a_M = 1 + O_c(1/\ln M)$ and

$$\mathbb{E}|\Delta| \le \frac{2\,\mathbb{E}R_k}{s_k} \le e^{-\mu_k}s_k = O_c\left(\sqrt{\frac{\ln M}{M}}\right),$$

using $\mathbb{E}R_k \le \frac{1}{2}e^{-\mu_k}s_k^2$, $e^{-\mu_k} = O(\ln M)$, and $s_k = O((M\ln M)^{-1/2})$. Choose $\varepsilon := (\ln M/M)^{1/4}$ in (28); Markov's inequality gives $\mathbb{P}(|\Delta| > \varepsilon) \le \mathbb{E}|\Delta|/\varepsilon = O_c((\ln M/M)^{1/4})$. Applying (28) and then (29) with $a = a_M$,

$$\begin{aligned}\sup_x \left|\mathbb{P}\left(\frac{Y_k - \rho_k(M)}{P_M s_k} \le x\right) - \Phi(x)\right| &= O_c\big((\ln M)^{-2/5}\big) + O_c\big((\ln M/M)^{1/4}\big) + O_c\big(1/\ln M\big) \\ &= O_c\big((\ln M)^{-2/5}\big).\end{aligned}$$

A final application of (29) with $a = r_M$ replaces $P_M s_k$ by $P_M\sqrt{B(c)/(M\ln M)}$, completing the proof. □

□

9 Numerical Evidence

The theorems in Sections 3–8 are proved rigorously and require no numerical verification. The numerics below are included to illustrate finite-M behaviour, and for Theorem 2 to document that descriptive diagnostics of Gaussianity are consistent with the proof. The variance calculation evaluates the exact finite formula in extended-precision floating-point arithmetic, with an independent 80-decimal arbitrary-precision spot check. Monte Carlo sampling uses NumPy's default PRNG, freshly initialised with the fixed seed 2026 for each panel.

Theorem 1. The rescaled variance $M\ln M \cdot s_k^2$ is evaluated deterministically from the exact finite formula $s_k^2 = V_1 + V_2$ via direct summation over $\mathcal{P}_M$ and $\mathcal{P}_M \times \mathcal{P}_M$. Figure 1 plots $M\ln M \cdot s_k^2$ versus M at $k = \lfloor cM \rfloor$ for $c \in \{0.5, 1, 2, 3\}$ and M up to 3200. Table 1 provides representative numerical values. Convergence to $B(c)$ is slow and visibly nonmonotone at finite M: at $M = 3200$, the rescaled variance agrees with $B(c)$ to within approximately 2% across $c \in [0.5, 3]$. The off-diagonal

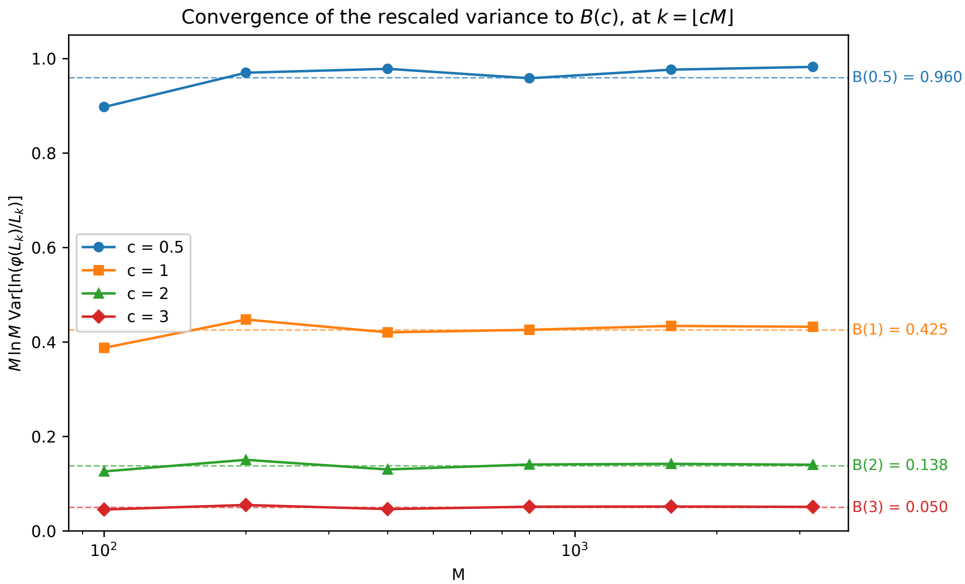

Figure 1: Rescaled variance $M \ln M \cdot s_k^2$ versus M at $k = \lfloor cM \rfloor$ for $c \in \{0.5, 1, 2, 3\}$. Values evaluated from the exact finite formula $s_k^2 = V_1 + V_2$ using extended-precision arithmetic. Dashed horizontal lines indicate the limit $B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$ predicted by Theorem 1.

Table 1: Deterministic extended-precision evaluation of $M \ln M \cdot s_k^2$ at $k = \lfloor cM \rfloor$ across $c \in \{0.5, 1, 2, 3\}$ and $M \in \{100, 200, 400, 800, 1600, 3200\}$, alongside the limiting value $B(c)$ from Theorem 1. Values computed via $s_k^2 = V_1 + V_2$ by direct summation; no Monte Carlo.

M	$c = 0.5$	$c = 1.0$	$c = 2.0$	$c = 3.0$
100	0.8977	0.3877	0.1261	0.0454
200	0.9702	0.4478	0.1506	0.0550
400	0.9785	0.4205	0.1303	0.0464
800	0.9585	0.4259	0.1407	0.0514
1600	0.9767	0.4342	0.1422	0.0518
3200	0.9826	0.4323	0.1403	0.0510
$B(c)$	0.9595	0.4255	0.1379	0.0499

contribution V_2 is always negative (consistent with Remark 10) and an order of magnitude smaller than V_1.

Theorem 2. Figure 2 displays the empirical distribution of the standardised coprime density $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$ at $k = M$ for $M \in \{100, 300, 1000\}$, where $\hat{\rho}_k, \hat{\sigma}_k$ are the sample mean and sample standard deviation. Sample sizes are $N = 3000$ at $M = 100$ and $M = 300$; $N = 1000$ at $M = 1000$. Sample skewness decreases from 0.61 at $M = 100$ through 0.27 at $M = 300$ to 0.16 at $M = 1000$; excess kurtosis (Fisher convention) decreases from 0.58 at $M = 100$ through 0.12 at $M = 300$ to -0.17 at $M = 1000$. These are descriptive diagnostics rather than statistical tests; they are consistent with the proof above. The observed skewness decays roughly by a factor 2 from $M = 100$ to $M = 300$ and again to $M = 1000$, appreciably faster than the $(\ln M)^{-2/5}$ guarantee of Theorem 3 (which changes by less than 20% across this range), consistent with that rate not being sharp (Section 10).

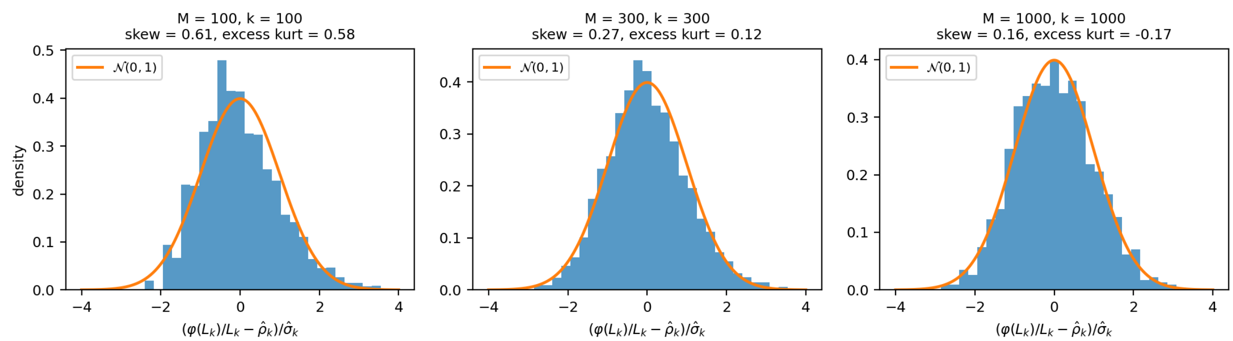

Figure 2: Empirical distribution of the standardised coprime density $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$ at $k = M$, for three values of M. Solid curve: standard normal density. Sample sizes N and Fisher-convention skewness/excess kurtosis reported in panel titles. Histograms use 35 bins in $[-4, 4]$ and are normalised to density.

10 Discussion

Why the increment bound needs both ingredients. The trivial Mertens-type bound $|D_i| \leq 2\Lambda(M) = O(\log\log M)$ is too large to feed into a Lyapunov condition: $(\log\log M)^2 \cdot M \cdot \ln M$ grows, not shrinks. A bound using only the coupon-collector estimate (restricting to primes with $m_p < (\log M)^2$, ignoring the sieve) yields $O(\log\log M)$ contributions per coordinate and is likewise insufficient. A bound using only the sieve-geometry (at most one contributing prime divides x) without the coupon-collector filter allows primes with very small q_p (i.e. very large m_p) that contribute as $O(1/\pi_p) = O(\log\log M)$ to the worst-case $|S_i(x)|$. Both estimates together reduce $|S_i(x)|$ to $(\log M)^2/(M-1)$, with the mean-correction $|C_i|$ contributing an additional $O((\log M)^3/M)$. The resulting uniform bound $\max_i |D_i| = O((\log M)^3/M)$ on Ω^* is what the Hall–Heyde hypotheses need.

Finite-M convergence in Theorem 1. Proposition 12 upgrades the bare convergence of Theorem 1 to the quantitative rate $M \ln M \cdot s_k^2 = B(c) + O_c(1/\ln M)$, obtained by replacing the pointwise prime number theorem in Lemma 7 by the effective Rosser–Schoenfeld bounds. This is consistent with the numerics of Table 1: at $M = 3200$ one has $1/\ln M \approx 0.12$, comfortably accommodating the 2–5% residuals observed there, and it explains the slow, mildly oscillatory approach without appeal to a sharper unproven rate. Whether the true rate is $O(1/\ln M)$ or smaller (the second-order companion expansion of [15] suggests the diagonal contributes a genuine $1/\ln M$ term) is left open.

The Berry–Esseen exponent. The rate $(\ln M)^{-2/5}$ of Theorem 3 is not expected to be sharp. In the Heyde–Brown bound (26) the fourth-moment term is $O_c((\ln M)^7/M)$, negligible against the conditional-variance term $\|V_k^2 - 1\|_{L^2}^2 = O_c((\ln M)^{-2})$; the exponent $2/5 = 2 \cdot \frac{1}{5}$ is thus entirely determined by the L^2 concentration scale of the predictable quadratic variation from Theorem 19, raised to the power $1/(3+2\delta) = 1/5$ at $\delta = 1$. The bottleneck is therefore the Minkowski bound $\mathrm{Var}(\sum_i U_i) = O_c(M^{-2}(\ln M)^{-4})$: any improvement to $O_c(M^{-2-\eta})$ for some $\eta > 0$ would, with no other change to the argument, replace the logarithmic rate by a power $M^{-\eta'}$. The exponent $1/(3+2\delta)$ in (26) is known to be unimprovable for general martingales [10], so a genuinely faster rate would have to come from the arithmetic structure rather than from a better abstract inequality. For comparison, the Erdős–Kac theorem attains the optimal rate $(\log\log n)^{-1/2}$ (Rényi–Turán [11]); the doubly-logarithmic scale there and the singly-logarithmic scale here both reflect that the relevant variance is carried by a slowly growing number of prime-indexed degrees of freedom.

General distributions. The Doob decomposition (Theorem 13) generalises verbatim to any distribution on $\{2, \ldots, M\}$, with $\xi_p^{(i)}$ redefined as $\mathbf{1}\{p \mid X_i\} - \mathbb{P}(p \mid X)$. The increment bound uses the uniformity of the X_i in two places: the coupon-collector estimate (which goes through if $\mathbb{P}(p \mid X) \gtrsim m_p/M$ uniformly) and the sieve-geometric observation (which is distribution-independent once levels are defined by q_p). The geometric disjointness in Proposition 6 is distribution-independent for any law supported on $\{2, \ldots, M\}$: no support point is divisible by both p and q when $pq > M$. What is specific to the uniform law is the explicit formula $\pi_p = m_p/(M-1)$ and the resulting level structure; a non-uniform extension would require distribution-specific estimates for these probabilities and covariances. Extension to product distributions close to uniform appears feasible; substantial departures from uniformity would change the limit $B(c)$.

Comparison with fixed-k CLT results. Buraczewski–Iksanov–Marynych [3] established a CLT for $\log \mathrm{lcm}$ of a uniform m-tuple on $\{1, \ldots, n\}$ with m fixed as $n \to \infty$. Our result is not comparable in regime: they study $\log \mathrm{lcm}$ itself, with k fixed, while we study $\log(\varphi(L_k)/L_k)$ with $k = \lfloor cM \rfloor$ and $M \to \infty$. The natural common ground — a CLT that interpolates between the two regimes, for instance $k = f(M)M$ with $f \to 0$ slowly — is not addressed here.

11 Conclusion

We have established, in the joint regime $k = \lfloor cM \rfloor$, $M \to \infty$, fixed $c > 0$, the variance asymptotic $M \ln M \cdot s_k^2 \to B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$ (Theorem 1) and a central limit theorem for the standardised log coprime density (Theorem 2). The variance theorem is proved by a direct diagonal / off-diagonal decomposition, with the off-diagonal term controlled by an exact covariance identity (Lemma 8) valid for prime pairs (p, q) with $pq > M$. The CLT is proved via the Hall–Heyde martingale central limit theorem applied to the Doob filtration. Its two nontrivial ingredients are a uniform increment bound combining coupon-collector concentration with a sieve-geometric observation (Theorem 14), and a conditional-variance concentration resting on per-prime moment bounds, the covariance estimate of Lemma 17, and an L^2 Minkowski bound that controls all prime-pair cross terms. Together with the first-order results of the companion paper [15], these give a two-moment asymptotic theory for the coprime density in this regime. A third result (Theorem 3) makes the CLT quantitative: the Kolmogorov distance to normality is $O_c((\ln M)^{-2/5})$, via the Heyde–Brown martingale inequality together with the quantitative variance rate of Proposition 12. The exponent is limited only by the L^2 concentration scale of the predictable quadratic variation, and would improve to a power of M under any polynomial sharpening of that scale.

Data and code availability. The numerical code, fixed random seed, generated figures, and machine-readable outputs underlying Table 1 and Figures 1–2 are openly available at

https://github.com/Wakid90/Variance-and-Central-Limit-Theorem-for-the-Log-Coprime-Density-of-Random-LCMs

The repository includes a one-command reproduction script and an arbitrary-precision validation mode for the exact variance formula.

References

[1] G. Alsmeyer, Z. Kabluchko, A. Marynych, *Limit theorems for the least common multiple of a random set of integers*, Trans. Amer. Math. Soc. 372 (2019), 4585–4603.

[2] A. Bostan, A. Marynych, K. Raschel, *On the least common multiple of several random integers*, J. Number Theory 204 (2019), 113–133.

[3] D. Buraczewski, A. Iksanov, A. Marynych, *Central limit theorem for the least common multiple of a uniformly sampled m-tuple of integers*, J. Number Theory 233 (2022), 301–336.

[4] J. Cilleruelo, J. Rué, P. Šarka, A. Zumalacárregui, *The least common multiple of random sets of positive integers*, J. Number Theory 144 (2014), 92–104.

[5] P. Hall, C. C. Heyde, *Martingale Limit Theory and Its Application*, Academic Press, New York, 1980.

[6] C. C. Heyde, B. M. Brown, *On the departure from normality of a certain class of martingales*, Ann. Math. Statist. 41 (1970), 2161–2165.

[7] T. Hilberdink, L. Tóth, *On the average value of the least common multiple of k positive integers*, J. Number Theory 169 (2016), 327–341.

[8] A. Iksanov, A. Marynych, K. Raschel, *Asymptotics of arithmetic functions of GCD and LCM of random integers in hyperbolic regions*, Results Math. 77 (2022), article 165.

[9] Z. Kabluchko, A. Marynych, K. Raschel, *Multivariate multiplicative functions of uniform random vectors in large integer domains*, Results Math. 78 (2023), article 201.

[10] J.-C. Mourrat, *On the rate of convergence in the martingale central limit theorem*, Bernoulli 19 (2013), 633–645.

[11] A. Rényi, P. Turán, *On a theorem of Erdős–Kac*, Acta Arith. 4 (1958), 71–84.

[12] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. 6 (1962), 64–94.

[13] C. Sanna, *On the least common multiple of random q-integers*, Res. Number Theory 7 (2021), article 16.

[14] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Graduate Studies in Mathematics 163, Amer. Math. Soc., 2015.

[15] M. Wakid, *On the Expected Coprime Density of the Least Common Multiple of Random Integers*, Zenodo (2026), doi:10.5281/zenodo.21325880.